FEDERICO DI GENNARO

PhD Student in AI at ETH Zürich

@ fdigennaro@ethz.ch 𝕏 fededg__ FedericoDiGennaro in Federico Di Gennaro

RESEARCH VISION

My broad research interests focus on how AI agents can make reliable decisions in open-ended environments, where they need adapts over time their behavior or solutions to solve complex tasks.

EDUCATION

ETH Zürich — Oct. 2025 – present
Ph.D. in Computer Science, Statistical Machine Learning Group — Zürich, Switzerland
Advisors: Prof. Yang and Prof. Krause.

UC Berkeley — Feb. 2025 – Jun. 2025
Visiting Student Researcher, Department of Statistics — Berkeley, United States
Efficient Logistic Regression with Mixture of Sigmoids (AISTATS 2026).
Advisor: Prof. Zhivotovskiy.

Università di Milano — Sept. 2024 – Jan. 2025
Visiting Student Researcher — Milan, Italy
Instance-Dependent Regret Bounds for Nonstochastic Linear Partial Monitoring (NeurIPS 2025).
Advisor: Prof. Cesa-Bianchi.

EPFL — Sept. 2022 – Jun. 2025
M.Sc. in Applied Mathematics, Minor in Data Science — Lausanne, Switzerland
Advisor: Prof. Lénaïc Chizat.
Research Assistant, Chair of DOLA: Stein's Unbiased Risk Estimate for explicitly and iteratively regularized estimators.
Teaching Assistant: Applied Data Analysis (CS 401).

Politecnico di Milano — Sept. 2019 – Sept. 2022
B.Sc. in Mathematical Engineering — Milan, Italy
Exchange semester: Universidad Autónoma de Madrid.

EXPERIENCE

AXA — Feb. 2024 – Aug. 2024
Research Scientist Intern — Paris, France
Research on fairness and interpretability for deep learning on tabular data in the ML Research team.
Advisors: Dr. Laugel and Dr. Grari.

Jane Street — Oct. 2023
Quantitative Trading Camp Participant — London, UK
Selected for a one-week quantitative trading training camp.

HONORS AND SCHOLARSHIPS

Hasler-Stiftung Foundation Scholarship (2024). $9,000 for the research visit to UC Berkeley.
Politecnico di Milano Merit Scholarship (2021). Tuition fees and exchange semester support.
Ruffini's Cup (2019). First place, team Mathematical Olympiad.
Italian Mathematical Olympiad, Team National Finalist (2017–2019). Three-time national finalist.

LANGUAGES

Italian (native) | English (C1) | Spanish (B1) | French (A2)

PUBLICATIONS

[1] **Progressive Misalignment Can Emerge from Benign Recursive Self-Distillation**.
K. Abdel Sadek*, **F. Di Gennaro***, F. Yang. *Preprint*, 2026.

[2] **LLM-to-Flow Distillation: Teaching Natural Language Goals to Scientific Generators**.
R. De Santi*, **F. Di Gennaro***, A. Gnazzo*, S. Tang, P. Chatterjee, F. Yang, İ. İ. Ceylan, M. M. Bronstein, A. Krause. *Preprint*, 2026.

[3] **Fast Unlearning at Scale via Margin Self-Correction**.
F. Di Gennaro, A. Shevchenko, F. Yang. *Preprint*, 2026.

[4] **Hedging on the Frontier: Learning New Tasks with Few Samples**.
T. Wegel, **F. Di Gennaro**, G. So, F. Yang. *ICML*, 2026. **Spotlight** (top 2.2%).

[5] **Efficient Logistic Regression with Mixture of Sigmoids**.
F. Di Gennaro, S. Chakraborty, N. Zhivotovskiy. *AISTATS*, 2026.

[6] **Instance-Dependent Regret Bounds for Nonstochastic Linear Partial Monitoring**.
F. Di Gennaro*, K. Eldowa*, N. Cesa-Bianchi. *NeurIPS*, 2025.

[7] **Controlled Model Debiasing through Minimal and Interpretable Updates**.
F. Di Gennaro, T. Laugel, V. Grari, M. Detyniecki. *Transactions on Machine Learning Research (TMLR)*.